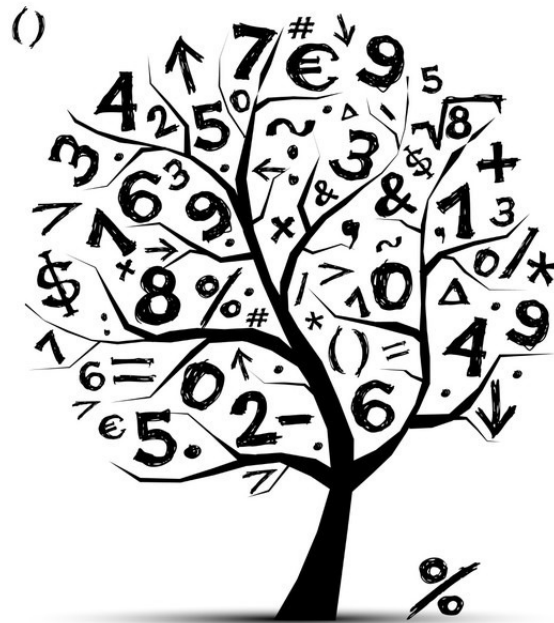


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Exercises: absolute value



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Exercise 1

Using the definition of absolute value, compute:

$$\text{a) } |-3| + |2-7| - |-4| \left| \frac{1}{2} \right|, \quad \text{b) } |3-\pi|, \quad \text{c) } \left| \frac{1}{3} - \frac{1}{2} \right|$$

a) Since $-3 < 0$, $2-7 = -5 < 0$, $-4 < 0$ and $\frac{1}{2} \geq 0$:

$$|-3| + |2-7| - |-4| \left| \frac{1}{2} \right| = 3 + 5 - 4 \cdot \frac{1}{2} = 6$$

b) Since $\pi > 3$, we have $3-\pi < 0$ and then $|3-\pi| = -(3-\pi) = \pi-3$.

c) Since $\frac{1}{3} - \frac{1}{2} = -\frac{1}{6} < 0$, we have $\left| \frac{1}{3} - \frac{1}{2} \right| = \frac{1}{6}$.

Exercise 2

Prove that for every real number b we have:

$$\text{a) } |b| \geq 0 \text{ and } |b| = 0 \iff b = 0, \quad \text{b) } -|b| \leq b \leq |b|, \quad \text{c) } |b|^2 = b^2$$

We distinguish the two cases of the definition of absolute value.

- If $b \geq 0$, then $|b| = b \geq 0$; moreover $|b| = 0$ if and only if $b = 0$; we have $-|b| = -b \leq 0 \leq b = |b|$; finally $|b|^2 = b^2$.
- If $b < 0$, then $|b| = -b > 0$ (in particular $|b| \neq 0$); we have $-|b| = b < 0 < -b = |b|$; finally $|b|^2 = (-b)^2 = b^2$.

In both cases a), b) and c) hold.

Exercise 3

Find all real numbers x such that:

$$\text{a) } |x-2| < 3, \quad \text{b) } |2x+1| \leq 5, \quad \text{c) } |x-4| \geq 1$$

a) From $|a| < \varepsilon \iff -\varepsilon < a < \varepsilon$ with $a = x - 2$ and $\varepsilon = 3$:

$$-3 < x - 2 < 3 \iff -1 < x < 5$$

b) From $|a| \leq \varepsilon \iff -\varepsilon \leq a \leq \varepsilon$ with $a = 2x + 1$ and $\varepsilon = 5$:

$$-5 \leq 2x + 1 \leq 5 \iff -6 \leq 2x \leq 4 \iff -3 \leq x \leq 2$$

c) The inequality $|x - 4| \geq 1$ is false exactly when $|x - 4| < 1$, that is when $-1 < x - 4 < 1$, i.e., $3 < x < 5$. Then the solutions are:

$$x \leq 3 \text{ or } x \geq 5$$

Exercise 4

Find all real numbers x such that $|x - 1| = |x + 3|$.

Two real numbers have the same absolute value if and only if they are equal or opposite. Then:

- $x - 1 = x + 3$ gives $-1 = 3$, which is impossible;
- $x - 1 = -(x + 3)$ gives $2x = -2$, that is $x = -1$.

The only solution is $x = -1$. Check: $|-1 - 1| = 2 = |-1 + 3|$.

Exercise 5

Find all real numbers x such that $|x| + |x - 2| = 4$.

The signs of x and $x - 2$ change at $x = 0$ and $x = 2$, so we distinguish three cases:

- $x < 0$: $|x| = -x$ and $|x - 2| = 2 - x$, then $-2x + 2 = 4$, i.e., $x = -1$, which satisfies $x < 0$;
- $0 \leq x < 2$: $|x| = x$ and $|x - 2| = 2 - x$, then $x + 2 - x = 2 = 4$, which is impossible;
- $x \geq 2$: $|x| = x$ and $|x - 2| = x - 2$, then $2x - 2 = 4$, i.e., $x = 3$, which satisfies $x \geq 2$.

The solutions are $x = -1$ and $x = 3$.

Exercise 6

Prove that for every pair of real numbers b and c :

$$|b + c| = |b| + |c| \iff bc \geq 0$$

Both $|b + c|$ and $|b| + |c|$ are non-negative, and two non-negative numbers are equal if and only if their squares are equal. Using Exercise 2 c) and $|bc| = |b||c|$, we have:

$$|b + c|^2 = (b + c)^2 = b^2 + 2bc + c^2, \quad (|b| + |c|)^2 = b^2 + 2|bc| + c^2$$

Then $|b + c| = |b| + |c|$ if and only if $bc = |bc|$, that is (by definition of absolute value) if and only if $bc \geq 0$.

Exercise 7

Verify the triangle inequality $|b + c| \leq |b| + |c|$ and the reverse triangle inequality $||b| - |c|| \leq |b - c|$ for $b = 5$ and $c = -3$.

We have $|b| = 5$ and $|c| = 3$.

- Triangle inequality: $|b + c| = |5 - 3| = 2 \leq 8 = |b| + |c|$.
- Reverse triangle inequality: $||b| - |c|| = |5 - 3| = 2 \leq 8 = |5 - (-3)| = |b - c|$.

The triangle inequality holds with the strict inequality, in agreement with Exercise 6, since $bc = -15 < 0$.

Exercise 8

Let x be a real number such that $|x - 3| \leq 1$. Prove that:

$$\text{a) } |x| \leq 4, \quad \text{b) } |x^2 - 9| \leq 7$$

and show that the bound in b) cannot be improved.

- a) From the form $|g| \leq |g - h| + |h|$ of the triangle inequality, with $g = x$ and $h = 3$:

$$|x| \leq |x - 3| + |3| \leq 1 + 3 = 4$$

- b) We have $x^2 - 9 = (x - 3)(x + 3)$ and then $|x^2 - 9| = |x - 3||x + 3|$. From the triangle inequality:

$$|x + 3| = |(x - 3) + 6| \leq |x - 3| + 6 \leq 7$$

and therefore $|x^2 - 9| = |x - 3||x + 3| \leq 1 \cdot 7 = 7$.

The bound cannot be improved: for $x = 4$ (which satisfies $|4 - 3| = 1$) we have $|16 - 9| = 7$.